

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Duration :60 Minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
- Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. **Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Answer:

- A retail company generates data from physical stores, online shopping, inventory systems, and customer activities. The company needs real-time inventory tracking, demand forecasting, and personalized marketing.
 - Apache Kafka can be used to collect real-time sales, inventory, and customer activity data. HDFS or cloud storage such as Amazon S3 can store large volumes of historical data. Structured sales and inventory data can be stored in relational databases, while customer browsing and clickstream data can be stored in semi-structured formats such as JSON or Parquet.
 - Apache Spark can process large historical datasets, while Spark Streaming or Apache Flink can process real-time inventory updates. Machine learning algorithms such as Random Forest, XGBoost, and LSTM can be used for demand forecasting.
 - For personalized marketing, collaborative filtering and content-based recommendation techniques can analyze customer purchase and browsing history. Power BI or Tableau can provide dashboards showing sales, inventory levels, predicted demand, and customer behavior.
 - Thus, the proposed architecture provides real-time inventory visibility, accurate demand forecasting, and personalized customer recommendations.
2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

Answer:

- A social media platform generates billions of posts, comments, likes, shares, and login activities. The main objective is to detect fake accounts and misinformation quickly.
- Apache Kafka can be used as the real-time data ingestion system. It collects user activities and divides them into partitions so that large volumes of data can be processed in parallel.
- Apache Flink or Spark Streaming can process these events in real time. Features such as account age, posting frequency, number of followers, sharing patterns, and unusual login locations can be extracted.
- Machine learning algorithms such as Random Forest, XGBoost, Logistic Regression, and Neural Networks can classify suspicious accounts. For misinformation detection, Natural Language Processing (NLP) can analyze the content of posts.
- Historical data can be stored in a data lake using HDFS or cloud storage. Graph analytics can also identify groups of fake accounts that repeatedly interact or share similar content.

- Kafka replication, partitioning, and distributed processing provide scalability and fault tolerance. Alerts can be generated when suspicious activities are detected.
- Thus, Kafka, Flink/Spark Streaming, machine learning, NLP, and graph analytics can provide fast and scalable social media fraud detection.

3. **Government Data Platform Problem** - A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Answer:

- A government agency needs to combine census, economic, and geospatial data from different departments. The major requirements are data sharing, interoperability, security, privacy, and proper governance.
- Data can be collected from different departments using APIs, ETL/ELT tools, and data integration systems. A centralized data lake can store raw data, while a data warehouse can store cleaned and structured data for reporting.
- Different departments may use different formats and database systems. Therefore, common data formats, metadata standards, APIs, and unique identifiers should be used to improve interoperability.
- A strong data governance strategy should define data ownership, data quality, access policies, metadata, retention rules, and data-sharing procedures. A data catalog can help departments find and understand available datasets.
- Sensitive citizen data should be protected using encryption, authentication, role-based access control, data masking, and audit logs. Geospatial data can be managed using systems such as PostgreSQL with PostGIS.
- Finally, Apache Spark can perform large-scale analytics, while Power BI or Tableau can provide authorized users with dashboards and reports.
- Thus, proper governance, interoperability, security, and centralized data management are essential for a national government data platform.

4. **Banking Fraud Detection Problem** - A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

Answer:

- Banks process millions of transactions every minute and need to identify fraudulent transactions with very low latency. The system must analyze both historical and real-time data.

- Apache Kafka can collect transaction events such as transaction amount, location, merchant, device, and time. Kafka partitions allow transactions to be processed simultaneously.
- Apache Flink or Spark Streaming can analyze transactions in real time. The current transaction can be compared with the customer's previous behavior to identify unusual activities.
- Machine learning algorithms such as Logistic Regression, Random Forest, XGBoost, and Isolation Forest can be used. Supervised learning can detect known fraud patterns, while anomaly detection can identify unusual transactions.
- Historical transaction data can be stored in a data lake or data warehouse and processed using Apache Spark for model training. A fast cache such as Redis can provide frequently required customer information.
- Encryption, authentication, access control, replication, and monitoring should be implemented for security and reliability.
- Thus, a Kafka-based streaming pipeline combined with machine learning can provide fast and accurate real-time fraud detection.

5. **Agriculture Smart Farming Problem** - A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Answer:

- Smart farming systems collect data from soil sensors, weather stations, drones, satellites, and farm equipment. The objective is to improve crop yield and reduce the unnecessary use of water and fertilizers.
- Sensors can collect soil moisture, soil pH, temperature, humidity, rainfall, and nutrient information. Drones can provide crop images and identify unhealthy areas.
- MQTT or Apache Kafka can collect data from distributed sensors. HDFS or cloud storage can store large volumes of historical sensor and drone data. Time-series databases can be used for continuous sensor readings.
- Apache Spark can process historical agricultural data, while Spark Streaming or Flink can process real-time sensor data.
- Machine learning algorithms such as Decision Tree, Random Forest, XGBoost, and LSTM can be used for crop yield prediction, disease detection, irrigation prediction, and weather-impact analysis.
- Thus, big data and machine learning can help farmers make better decisions, increase productivity, reduce costs, and use agricultural resources efficiently.